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Performance Improvement of YOLOv13n for Waste Detection Using Contrast-Limited Adaptive Histogram Equalization in a Web-Based System

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1. Introduction

Global solid waste generation reached approximately 2.01 billion tonnes in 2016 and is projected to surge to 3.40 billion tonnes by 2050[1], [2]. In Indonesia, annual waste generation has reached a critical level of approximately 25 million tons[3], with current data indicating that only 66% is managed effectively[3]. Traditional frameworks struggle to keep pace due to a reliance on manual sorting, which is not only inefficient and prone to categorization errors but also exposes workers to potential physical risks and unhygienic materials found in common waste streams[4], [5], [6]. Artificial Intelligence and computer vision have emerged as revolutionary tools for addressing global waste issues[7]. Deep learning models can automatically extract intricate features from visual data for high-precision classification[8]. Within the field of object detection, single-stage detectors such as You Only Look Once (YOLO) perform detection and classification in a single evaluation step. The YOLO family of algorithms has undergone significant evolution from early versions to high-performance architectures like YOLOv12[10]. Architectural innovations such as anchor-free detection and enhanced feature extraction have consistently improved model speed[11].

[1]Typeset names in 8 pt Times Roman. Use the footnote to indicate the present or permanent address of the author.
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Despite significant algorithmic advancements, detecting waste in complex real-world environments remains a major challenge[12]. Environmental factors such as lighting fluctuations can significantly alter the visibility of an object[13]. Poor illumination and shadows often lead to missed detections, particularly for small or overlapping items[14]. In urban surveillance or underwater scenes, low image contrast degrades the recognition accuracy of deep learning models[15]. Preprocessing via image enhancement is a fundamental strategy for addressing these environmental visibility constraints[4]. Standard steps such as resizing and pixel normalization ensure input consistency for neural networks[16]. Contrast-Limited Adaptive Histogram Equalization (CLAHE) offers a robust mechanism for highlighting features in low-contrast visual data[17]. Integrating robust preprocessing methods can provide the visual clarity necessary for high-precision sorting.[8]

This research explores the tools and technologies, ranging from open-source deep learning frameworks to web-based deployment platforms, that facilitate the automated management of waste. Algorithms serve as the core of these artificial intelligence systems, driving decision-making processes and enabling intelligent sorting behavior[7]. This paper aims to provide a performance evaluation of the techniques and challenges in waste detection using the latest YOLOv13n architecture. We delve into the underlying theories of feature extraction, explore the basis of image enhancement, and discuss the importance of local contrast optimization[7]. Additionally, we survey existing methodologies, ranging from traditional machine learning to deep learning models such as YOLOv5 and YOLOv8, and analyze their limitations in low-contrast scenarios [10]. By addressing these challenges and discussing the integration of Contrast-Limited Adaptive Histogram Equalization, the goal is to serve as a resource for researchers seeking to leverage artificial intelligence for smart city infrastructure[18]. We collected the journal papers from reputable digital databases and searched around 50 papers with keywords such as waste detection and YOLO to establish a theoretical baseline[1]. Next, we analyzed these papers to identify the research gap regarding the newest iterations of real-time detectors[1].

A primary innovation of this study is the performance evaluation of the YOLOv13n model, which represents a new iteration beyond the architectures documented in current literature[10]. This research introduces an original integration of Contrast-Limited Adaptive Histogram Equalization to enhance local contrast in low-visibility waste imagery [19]. By utilizing the nano-sized variant of the architecture, the study optimizes the system for high-speed processing on hardware with limited resources[20]. The proposed system provides a scalable web-based interface that supports modern urban waste management frameworks[21]. This integration facilitates the development of automated sorting machines and recycling bins powered by artificial intelligence[11]. Furthermore, the system helps mitigate health risks associated with the manual handling of sharp or contaminated recyclable materials[22]. This breakthrough provides a replicable methodology that can be integrated into diverse smart city infrastructures[23]. The insights gained from this study contribute to the development of more accurate and robust systems capable of managing urban litter effectively[1].

The next section discusses some relevant research in the field of waste detection using deep learning approaches. Section 3 discusses the methodology used in automated waste

detection. More specifically, information is given about the datasets used, the algorithms used, as well as the experimental steps performed. The results of the experiments conducted in waste detection using the YOLO algorithm are discussed in Sec. 4. From the experimental results and analysis conducted with the various algorithms, Sec. 5 describes the implications and limitations of the findings. Finally, Sec. 6 shows the conclusions and avenues for future research.

2. Related Work

2.1. *Deep learning in waste management system*

The integration of deep learning into waste management systems has emerged as a significant approach to addressing the growing challenge of global waste accumulation, marking a shift from conventional methods toward intelligent automation. Through computer vision techniques, these systems enable real-time detection, localization, and classification of various waste materials. Such approaches have been widely implemented in automated sorting facilities, waste-sorting robots, and smart bin applications that monitor fill levels and support waste separation at the source[24]. The main advantage of deep learning approaches over manual methods lies in their ability to substantially improve operational speed and efficiency, while reducing reliance on human labor that is prone to error and potential health risks from prolonged exposure to waste materials[25].

However, the practical deployment of waste detection systems remains a challenge. System performance is frequently affected by environmental variability, including substantial variations in illumination, unfavorable weather conditions, complex visual backgrounds, and waste objects that are partially occluded, overlapping, or physically deformed[26]. This indicates that although deep learning has become a crucial component in the modernization of waste management, a gap remains between high accuracy achieved in controlled environments and reliable performance in dynamic real world conditions, which continues to be a major challenge to address[24][27].

2.2. *Waste detection using object detection models*

2.2.1. *Transition from image classification to object detection*

The transition from manual waste sorting to automated intelligent systems has been accelerated by advances in Deep Learning, particularly through the use of Convolutional Neural Networks. Early studies primarily relied on image classification approaches that assigned a single label to an entire image. However, recent literature increasingly adopts object detection frameworks, as they enable not only waste categorization but also precise localization of multiple objects within complex real world scenes, which is essential for automated sorting and robotic manipulation[28].

2.2.2. *Single stage and two stage detection frameworks*

Within the domain of object detection, research has generally been divided into two-stage and single-stage detectors. Although two-stage approaches are often associated with high detection accuracy, a clear trend in the literature favors single-stage detectors, particularly the You Only Look Once family, for waste management applications. This preference is primarily driven by the need for fast inference and computational efficiency, which are critical for real time deployment in automated sorting systems and resource constrained environments[25], [29].

This preference is primarily driven by the computational efficiency and fast inference capability of single stage models, which are essential for real time deployment in resource constrained systems such as reverse vending machines, autonomous cleaning robots, and smart bins[28]. Existing studies consistently report that although two stage detectors can achieve slightly higher precision in certain scenarios, their higher latency limits their practicality for continuous and high-speed waste processing on conveyor belts or mobile monitoring platforms[30][31].

2.2.3. *Application scenarios in waste detection*

The application of object detection models in waste management has formed several research clusters. A prominent area concentrates on the separation of municipal solid waste and recyclables, with models designed to differentiate between organic and inorganic materials[32][33]. A further line of research emphasizes fine grained material classification, which involves distinguishing between visually similar waste materials and relies on the ability of detection models to capture subtle visual differences[34][35]. Furthermore, object detection has been increasingly adopted for environmental monitoring, including the detection of floating debris in river and marine environments, where background dynamics and illumination variability remain challenging[30].

2.2.4. *Limitations and open challenges*

Despite their extensive use, YOLO based detection models continue to face robustness limitations. Numerous studies report performance degradation when these models operate under complex and unconstrained real-world conditions[26][36]. Standard models frequently struggle with small waste objects, such as cigarette butts or bottle caps, which occupy minimal pixel areas and are easily obscured by cluttered backgrounds[25]. Additionally, challenges related to object occlusion, overlapping waste items, and variations in lighting conditions continue to reduce the reliability of automated detection systems [36].

These challenges highlight an existing research gap, as most studies focus on architectural modifications, while the potential of image enhancement and preprocessing techniques to address environmental variability and low contrast

remains underexplored[28][25]. Consequently, further investigation is required to develop integrated systems that combine lightweight detection models with robust image processing strategies to ensure high accuracy and efficiency in unconstrained waste management scenarios.

2.3. *Performance improvement through image enhancement*

Beyond environmental factors, degraded image quality has been widely recognized as a key contributor to reduced detection performance. Low contrast, uneven illumination, shadows, and motion blur can obscure critical visual cues, making it difficult for detection models to extract discriminative features, particularly in cluttered scenes or low light conditions[37]. To address these challenges, much of the existing literature emphasizes the use of data augmentation strategies during the training phase. These approaches seek to improve model generalization by introducing artificial variability, commonly through geometric transformations and photometric adjustments such as changes in brightness, exposure, and contrast, encouraging models to become more resilient to adverse lighting conditions [24]. In parallel, architectural enhancements such as attention mechanisms and multi scale feature fusion have been investigated to improve feature discrimination in cluttered scenes[26]. While these methods enhance model robustness, they fundamentally address the symptom via training complexity rather than correcting the input quality itself.

In contrast, the use of image enhancement techniques as a preprocessing step remains relatively underexplored within the broader waste detection literature. Although some domain specific studies have applied enhancement or restoration methods, particularly in specialized settings, reported outcomes vary considerably[30]. This variability suggests that the role of image enhancement in improving detection performance has not been systematically examined, especially in relation to lightweight object detection models operating under practical constraints.

3. Methodology

3.1. *Datasets*

The dataset used in this study is the WasteIn Dataset developed by Alam Rahmatulloh et al. in 2025[38]. It consists of waste images categorized into six classes, namely electronic, glass, metal, organic, paper, and plastic. The images are collected from various sources and include variations in lighting conditions, viewing angles, and backgrounds. Figure 1 presents sample images from the WasteIn Dataset, while Table 1 provides a summary of the dataset.

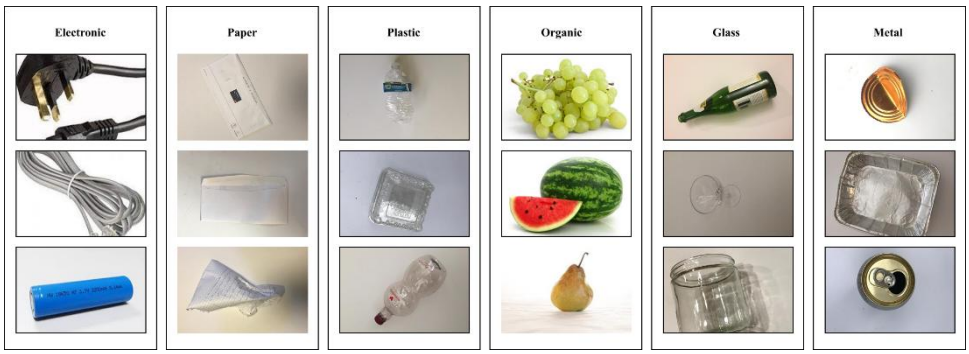

Fig. 1. Sample images from the WasteIn dataset[38].

Table 1. Description of WasteIn dataset.

Class	Number of Images
Electronic	200
Paper	200
Plastic	200
Organic	200
Glass	200
Metal	200

All images are used for training and testing an object detection model based on You Only Look Once (YOLO). In this study, the YOLOv13 nano variant is used due to its lightweight architecture. To examine the effect of image enhancement, two experimental settings are applied, namely using the original images and using images processed with Contrast-Limited Adaptive Histogram Equalization (CLAHE). In the second setting, all images are enhanced to improve contrast, allowing clearer representation of object features. The performance of both settings is then compared to evaluate the impact of CLAHE on detection performance. The CLAHE-enhanced version of the WasteIn Dataset generated in this study is publicly available at Enhanced Waste Dataset. Figure 2 illustrates sample images from the WasteIn Dataset that have been processed using CLAHE for contrast enhancement, while Table 2 provides a summary of the dataset.

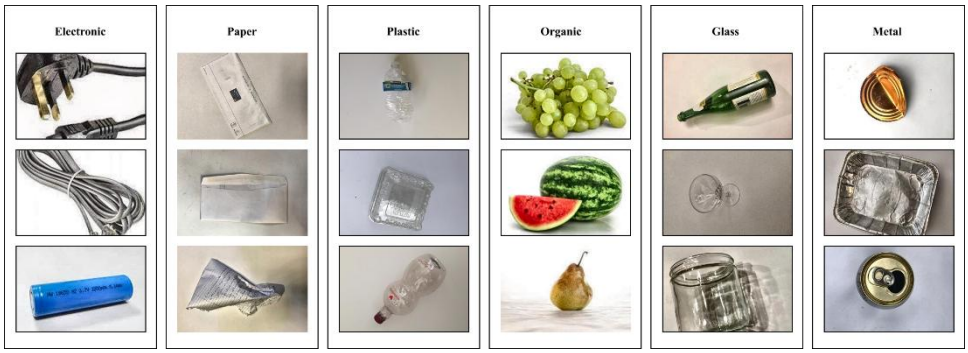

Fig. 2. Sample images after CLAHE preprocessing.

Table 2. Description of the CLAHE-Enhanced Images.

Class	Number of Images
Electronic	200
Paper	200
Plastic	200
Organic	200
Glass	200
Metal	200

3.2. *Experimental procedure*

The process consists of three main stages, starting from preprocessing, followed by model training, and ending with object detection. During preprocessing, the dataset is prepared, where an additional enhancement step using Contrast-Limited Adaptive Histogram Equalization (CLAHE) can be applied to improve image contrast. The training stage utilizes You Only Look Once version 13 (YOLOv13) with its nano variant to learn object features from both the original and enhanced images. In the final stage, the trained model is used to detect and classify waste objects into their respective categories. The overall workflow is presented in Fig. 3.

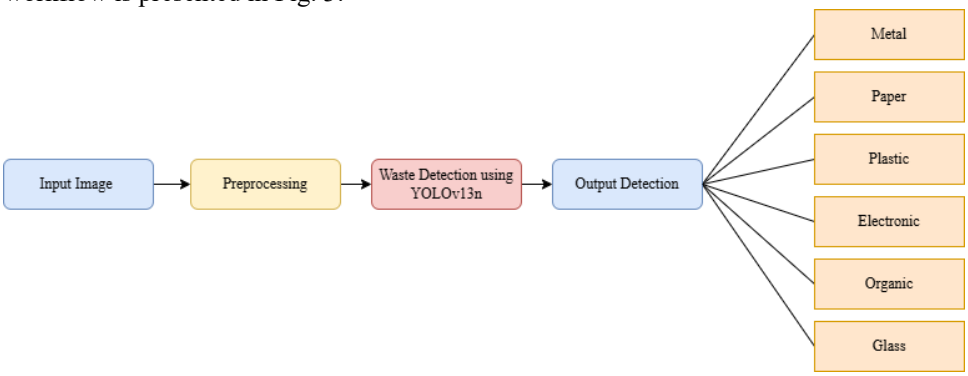

Fig. 3. Flowchart waste detection phases.

3.3. *Preprocessing*

The preprocessing phase focuses on spatial and contrast adjustments to ensure feature consistency across the dataset. Every image is resized to a uniform resolution of 640 x 640 pixels to maintain compatibility with the YOLOv13 nano architecture. To address inconsistent lighting conditions, the study implements a dual-stream strategy by including both original raw images and images enhanced through Contrast Limited Adaptive Histogram Equalization (CLAHE). This enhancement highlights object boundaries and improves local contrast, allowing the model to learn effectively from both natural and contrast-optimized visual data. CLAHE limits contrast amplification by applying a clip limit to each histogram tile, which prevents noise over-enhancement. The clip limit is defined as shown in Formula 1.

$$\beta = \frac{M}{N}(1 + \frac{\alpha}{100}(Smax - 1)) \quad (1)$$

From Formula 1, the explanations are C denotes the clip limit, M represents the total number of pixels within a tile, and N is the number of histogram bins. The term M/N indicates the average number of pixels per bin. Furthermore, α is the contrast control factor that determines the degree of enhancement, and Smax corresponds to the maximum gray level value in the image[39].

3.4. *Model*

3.4.1. *YOLOv13*

The architectural design of YOLOv13 represents a significant refinement in the efficiency of real-time object detection. Central to this evolution is the implementation of the Hypergraph-based Adaptive Correlation Enhancement (HyperACE) mechanism, which introduces a transformative approach to feature representation. Unlike previous iterations such as YOLOv11 and YOLOv12 that primarily relied on standard convolutional architectures or area-based self-attention mechanisms limited to pairwise correlations, YOLOv13 shifts the focus toward higher-order data relationships. By utilizing hypergraph computation, the HyperACE framework is capable of capturing global, multi-to-multi high-order correlations. This method effectively transcends the spatial constraints of traditional edge-based connections, allowing the model to aggregate complex contextual information more holistically across the entire feature map[40][41]. The model enhances perceptual capabilities by mapping multi-scale features to hypergraph vertices. Using a learnable participation matrix, it dynamically constructs hyperedges to exploit latent semantic relationships across diverse scales and locations. This adaptive approach allows for superior contextual representation in complex scenarios[40][42].

To make full use of these features, the You Only Look Once version 13 model introduces the Full Pipeline Aggregation and Distribution (FullPAD) approach. This mechanism distributes the correlation enriched features generated by HyperACE across all components of the network, including the backbone, neck, and detection head, allowing more detailed information flow and improving overall feature representation[40][41]. Furthermore, to preserve real-time performance, the You Only Look Once version 13 model replaces standard large-kernel convolutions with lightweight depthwise separable convolutions and incorporates efficient modules such as DS-C3k and DS-C3k2. This design reduces the number of parameters and computational cost, measured in floating point operations (FLOPs), while maintaining improved detection accuracy compared to earlier versions such as YOLOv11 and YOLOv12[40]. To provide a clearer understanding of how these mechanisms are

integrated, the overall structural organization of YOLOv13 is depicted in Fig.4, detailing the interplay between the feature extraction backbone and the advanced enhancement modules.

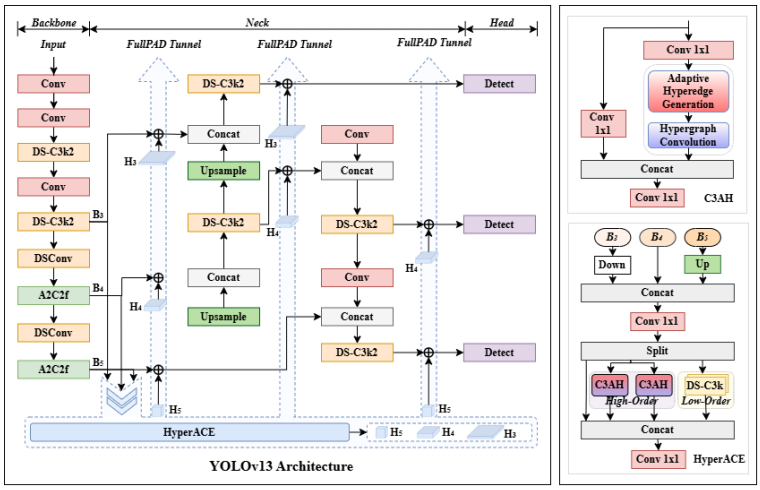

Fig. 4. YOLOv13 Architecture[42].

To optimize the model's parameters during the training phase, a multi-task loss function is employed to supervise the learning process. The total loss is calculated by aggregating three primary components and expressed as shown in Formula 2.

$$L = L_{box} + L_{obj} + L_{cls} \tag{2}$$

From Formula 2, the explanations are L_{box} represents the bounding box regression loss, which measures the precision of the predicted spatial coordinates compared to the ground truth. L_{obj} denotes the objectness loss, evaluating the model's confidence in determining whether a specific region contains an object or belongs to the background. L_{cls} signifies the classification loss, quantifying the accuracy of the predicted category labels for the detected objects.

3.4.2. *YOLOv13n*

YOLOv13 nano, introduced in 2025, is the most lightweight variant of the YOLOv13 architecture, specifically designed for high efficiency on edge devices with limited computational resources. Its key innovations include the Hypergraph-based Adaptive Correlation Enhancement (HyperACE) mechanism and the Full-Pipeline Aggregation-and-Distribution (FullPAD) paradigm, which enable the model to capture complex global feature correlations while maintaining low computational overhead compared to conventional self-attention approaches[41].

3.5. *Evaluation*

The performance of the YOLOv13-N model is quantitatively assessed using a comprehensive set of metrics to determine its effectiveness in waste detection. The primary evaluation utilizes mean Average Precision at a 0.5 IoU threshold (mAP50) to measure the model's ability to localize and classify objects with standard accuracy. To capture a more rigorous assessment of localization precision, the mAP50-95 metric is also employed, calculating the average precision across a range of Intersection over Union (IoU) thresholds from 0.5 to 0.95 with increments of 0.05. Furthermore, the reliability of the model is validated through Precision, which measures the ratio of correct detections against total positive forecasts, and Recall, which evaluates the model's sensitivity in identifying all existing ground truth instances. These combined metrics ensure a robust validation of the architecture's success in both object categorization and spatial coordinates[40].

4. Results and Discussion

4.1. Performance of YOLOv13n Based on CLAHE Enhancement

In this section, we review the research methods utilized to construct the object detection model for identifying various categories of waste. The You Only Look Once (YOLO) algorithm is regarded as a leading framework for smart waste management systems due to its high accuracy and real-time processing capabilities[2], [6]. Unlike traditional detectors, YOLO analyzes the entire image in a single pass to simultaneously predict bounding boxes and class probabilities[1], [22]. Figure 4 provides a detailed view of the YOLOv13n modules and its updated architectural backbone. Collecting the dataset, preparing the images with enhancement techniques, and training the network are essential parts of the work process[6].

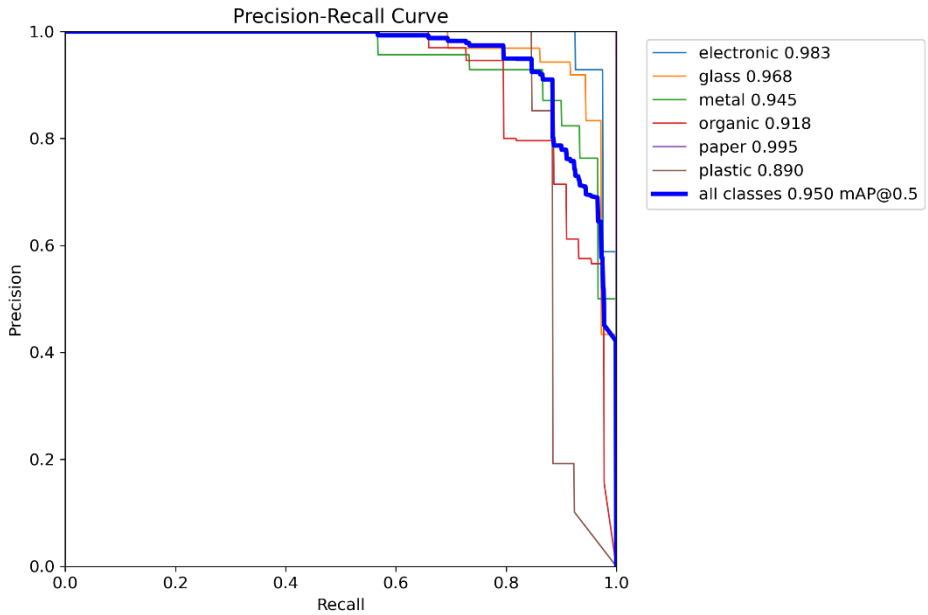

Fig. 5. Result of Precision-Recall Curve.

The Precision-Recall Curve, as shown in Fig. 5, illustrates the thorough visual evaluation of the model's identification performance. The mean Average Precision (mAP) metric is utilized to assess the effectiveness and reliability of these object detection systems[6], [43]. This study experimentally evaluated the detection performance of the YOLOv13n model using both original and enhanced datasets. Based on the test results, the enhanced model achieved a mean Average Precision at fifty percent intersection over union (mAP50) of zero point nine five. This value represents a measurable improvement over the original dataset, which reached a mAP50 of zero point nine four five.

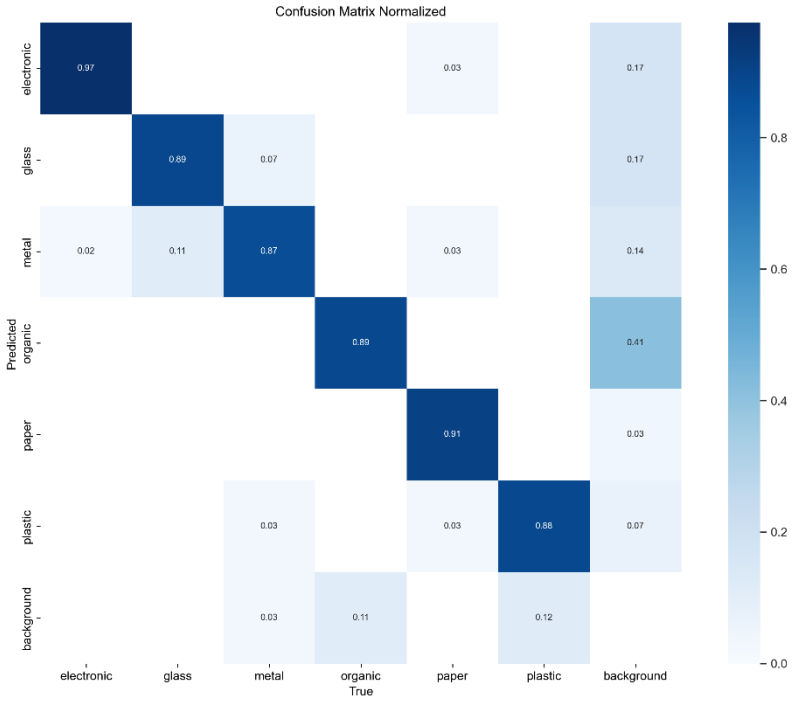

Fig. 6. Results of Testing the Confusion Matrix on The Model.

The accuracy of the proposed YOLOv13n model outputs was assessed using a confusion matrix[18], [19]. The projected proportions of each waste category are displayed in each of the confusion matrix's columns[6], [11]. Fig. 6 displays the actual proportions of each category in the data, as indicated by each row. As shown in Fig. 6, the latest prediction rates for the anticipated categories "electronic", "glass", "metal", "organic", "paper", and "plastic" are 97.6%, 94.1%, 82.8%, 93.8%, 100%, and 95.3%, in that order. It was found that the classification model's performance for the metal class was lower than others at 82.8%.

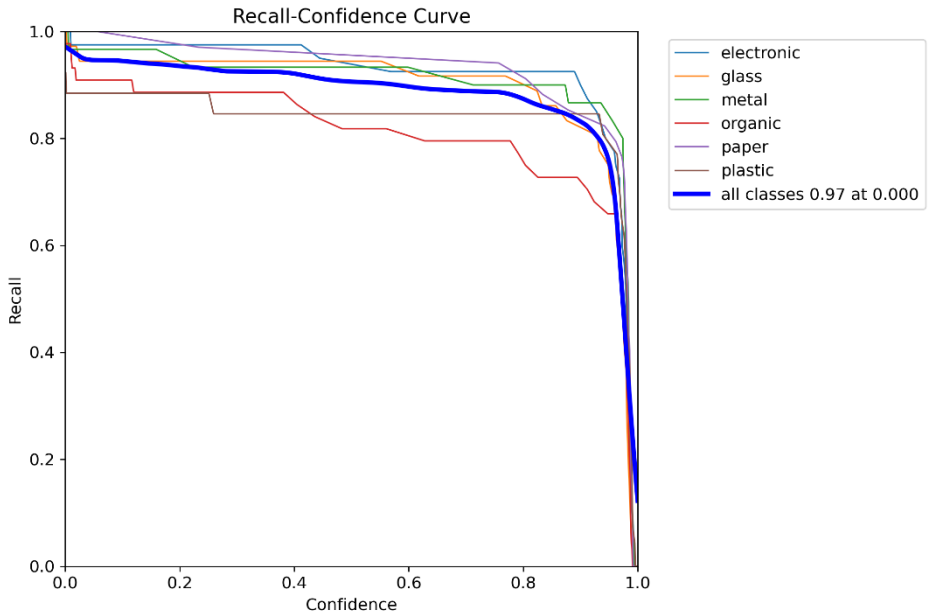

Fig. 7. Result of Recall-Confidence Curve.

The Recall-Confidence Curve, depicted in Fig. 7, evaluates the model's ability to identify all relevant instances at various confidence thresholds[19]. This visualization sheds light on the sensitivity of the detector across the six distinct waste categories[5], [19]. A high recall curve indicates that the model can effectively detect a wide range of objects with minimal omissions[4], [44]. Based on our evaluation, the enhanced YOLOv13n model achieved an overall recall of zero point eight eight seven. This reflects a high degree of comprehensiveness in identifying waste items even in challenging visual conditions[5], [6].

Environmental factors such as lighting conditions can significantly change the visibility of an object and alter its appearance[15], [45]. If image quality is poor due to low contrast, the accuracy of a deep learning model will be disrupted because the object to be analyzed becomes unclear[8], [46]. To enhance performance, this study integrated Contrast-Limited Adaptive Histogram Equalization (CLAHE) to address environmental visibility constraints in cluttered scenes[1], [47]. The results confirm that image enhancement provides the necessary visual clarity for high-precision waste sorting[4], [8].

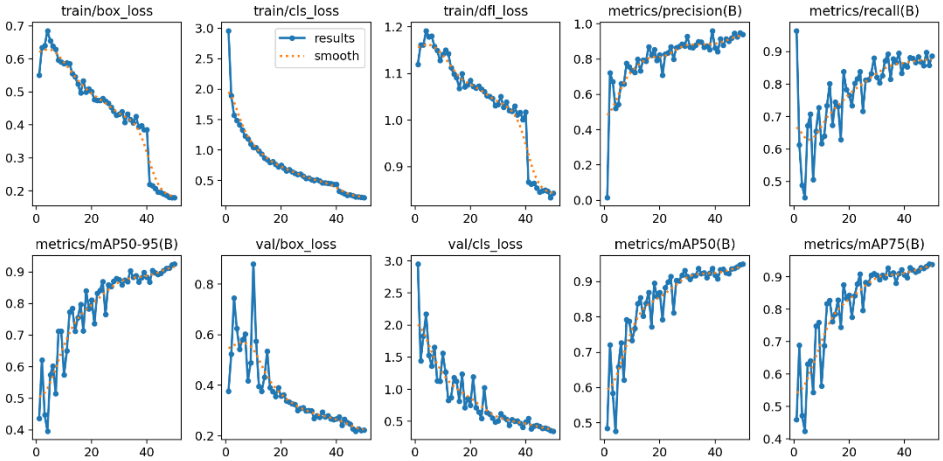

Fig. 8. The Graphical Representation of Loss Value Variations, Including the Box Loss.

A graphic depiction of the variations in loss values is also included in this study[7], [11]. The YOLO algorithm utilizes a loss function to correct errors in the center and bounding box coordinates of each prediction[16], [43]. Fig. 8 shows how the loss metrics consistently decreased as training progressed, indicating the model's improving accuracy. After fifty epochs, the loss values stabilized, showing that the training process achieved successful convergence.

Table 3
Training Results.

Class	Images	Labels	Precision	Recall	mAP@.5	mAP@.75	mAP@.5:.95
All	180	210	0.91	0.881	0.945	0.928	0.916
electronic	34	40	0.973	0.925	0.984	0.975	0.932
glass	36	36	0.917	0.994	0.97	0.97	0.964
metal	28	30	0.901	0.914	0.951	0.951	0.951
organic	27	44	0.81	0.705	0.868	0.788	0.771
paper	34	34	0.994	0.912	0.988	0.988	0.986
plastic	21	26	0.867	0.885	0.906	0.894	0.893

A testing method was used on the dataset to illustrate the scalability and performance of the YOLOv13n model further[13], [48]. Table 3 shows the outcomes of the performances throughout the training and validation phases. The model was trained in zero point seven zero two hours using 50 epochs. This hardware configuration utilizing the NVIDIA RTX 3050 6GB GPU provides the necessary computational strength for real-time monitoring applications.

4.2. Comparison with Related Works

Table 4
Comparison Results.

Research Reference	Model Architecture	mAP@0.5
Dipo et al. (2025)	YOLOv12	0.780
Sun and Zhang (2025)	YOLOv10	0.800
Arishi (2025)	YOLOv8-CBAM	0.895
Panmuang and Rodmorn (2024)	YOLOv5n	0.945
This Study	YOLOv13n + CLAHE	0.950

Based on the comparison results with previous related research as in Table 4, it can be seen that our proposed model is superior with an mAP@0.5 value of zero point nine five. The novelty and impact of this research produce a deep learning model that is very suitable for waste conditions in Indonesia, especially for single types of indoor waste. So, this model will be very ideal for indoor household waste management systems. High-precision identification is essential for transitioning from reactive waste management to proactive, technology-driven strategies.

This outcome is very much in accordance with our predictions and ideas that local contrast enhancement can outperform previous models by adding augmentation techniques to the dataset. Although the performance gain was incremental compared to the baseline, the system successfully outperforms previous studies utilizing earlier YOLO versions. The high mAP value highlights the overall competence of the architecture in distinguishing different waste categories[5]. Robust performance across multiple thresholds indicates that the model has reached a high level of performance in detecting garbage objects.

However, this research still has limitations and shortcomings related to the dataset, which can be seen in the lower accuracy value for the metal category. This challenge is consistent with previous findings where models struggle to distinguish objects with high visual similarities or reflective textures[10]. Classification errors are often likely due to similarities in lighting and specific shooting angles. This value still has the opportunity to be increased by optimizing the dataset and specifying a data collection protocol that is unique to this type of waste, especially in Indonesia.

The practical implication of this research is that the optimized YOLOv13n model can be used as a basis for further research. For example, the use of this model can be combined with Internet of Things-based tools or smart trash bins. Such integration facilitates the development of automated sorting machines that reduce human exposure to hazardous materials. By focusing on a web-based system, the proposed methodology provides a scalable solution for real-time environmental monitoring. The insights gained from this study contribute to the development of more accurate and robust systems capable of managing urban litter effectively.

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